

Tyler LaBonte

Senior Researcher

Microsoft Research AI Frontiers

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Research Interests

My research goal is to ensure **robust, grounded, and trustworthy multimodal agents** through foundational innovations in representation and generalization. Recently, I was a core member of the team behind [Phi-4-reasoning-vision](#), a 15B multimodal vision-language reasoning model with application to computer-use agents. My PhD work focused on theory and algorithms for robustness to distribution shift.

Education

GEORGIA INSTITUTE OF TECHNOLOGY

2021–2026

Ph.D., Machine Learning

Minor in Mathematics

Advisors: [Vidya Muthukumar](#) and [Jacob Abernethy](#)

UNIVERSITY OF SOUTHERN CALIFORNIA

2017–2021

B.S., Applied and Computational Mathematics, *magna cum laude*

Minor in Computer Science

Advisor: [Shaddin Dughmi](#)

Employment

1. MICROSOFT RESEARCH AI FRONTIERS

Remote

Senior Researcher

2026–

Future capabilities of multimodal agents.

2. MICROSOFT RESEARCH AI FRONTIERS

Redmond, WA

Machine Learning Research Intern

2025

Advisors: [Vibhav Vineet](#) and [Neel Joshi](#)

- Joined core development team of a 15 billion parameter multimodal vision-language model.
- Led integration effort, tool-use training, and evaluation for a multimodal computer-use agent application.
- Investigated grounding capabilities of our model on fine-grained visual reasoning tasks.

3. GOOGLE

Sunnyvale, CA

Machine Learning Research Intern

2023

Advisor: [Kun Lin](#)

- Developed techniques to leverage Gemini LLM to improve architecture-agnostic hardware-software code design.
- Synthesized chain-of-thought and few-shot prompting strategies to generalize to data-scarce applications.

4. MICROSOFT RESEARCH

Redmond, WA

Machine Learning Research Intern

2021–2022

Advisors: [Vibhav Vineet](#) and [Neel Joshi](#)

- Developed Transformer model for weakly supervised object detection via multiple instance learning.
- Integrated pipeline into production system, enabling rapid delivery of new Windows Action Center capability.

5. **GOOGLE X** Mountain View, CA
Machine Learning Research Intern 2020
Advisor: Daniel R. Silva
 - Invented CNN-LSTM for temporal identity preservation in multiple object tracking for computational agriculture.
 - Presented results to Google executives, who approved an FTE hire to deploy my research to production systems.
6. **SANDIA NATIONAL LABORATORIES** Albuquerque, NM
Machine Learning Research Intern 2019–2020
Advisors: Carianne Martinez and Scott A. Roberts
 - Invented novel Bayesian CNN deep learning architecture which scales to billion-voxel 3D segmentation volumes.
 - Enabled error bound calculation for physical properties of graphite electrodes and thermal protection systems.

Publications

An asterisk (*) denotes equal contribution. An ($\alpha\beta$) denotes alphabetical author ordering.

TECHNICAL REPORTS

1. [Phi-4-reasoning-vision-15B Technical Report](#)
($\alpha\beta$) Jyoti Aneja, Michael Harrison, Neel Joshi, Tyler LaBonte, John Langford, and Eduardo Salinas
Technical Report, 2026
Press coverage: [Microsoft Research Blog](#), [VentureBeat](#), [Forbes](#)
2. [We Know Where We Don't Know: 3D Bayesian CNNs for Credible Geometric Uncertainty](#)
Tyler LaBonte, Carianne Martinez, and Scott A. Roberts
Technical Report, 2019

CONFERENCE ARTICLES

1. [Task Shift: From Classification to Regression in Overparameterized Linear Models](#)
Tyler LaBonte*, Kuo-Wei Lai*, and Vidya Muthukumar
AISTATS 2025
INFORMS Applied Probability Society Conference 2025
IMS Workshop on Frontiers of Statistical Machine Learning 2025 (**top-10 award**)
2. [The Group Robustness is in the Details: Revisiting Finetuning under Spurious Correlations](#)
Tyler LaBonte, John C. Hill, Xincheng Zhang, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2024
3. [Towards Last-layer Retraining for Group Robustness with Fewer Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2023
4. [Scaling Novel Object Detection with Weakly Supervised Detection Transformers](#)
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
WACV 2023

JOURNAL ARTICLES

1. [On the Unreasonable Effectiveness of Last-layer Retraining](#)
John C. Hill, Tyler LaBonte, Xincheng Zhang, and Vidya Muthukumar
Transactions on Machine Learning Research, 2026.
2. [Student Misconceptions of Dynamic Programming: A Replication Study](#)
Michael Shindler, Natalia Pinpin, Mia Markovic, Frederick Reiber, Jee Hoon Kim, Giles Pierre Nunez Carlos, Mine Dogucu, Mark Hong, Michael Luu, Brian Anderson, Aaron Cote, Matthew Ferland, Palak Jain, Tyler

LaBonte, Leena Mathur, Ryan Moreno, and Ryan Sakuma.

Computer Science Education, 32(3):288–312, 2022

3. [Quantifying the Unknown Impact of Segmentation Uncertainty on Image-Based Simulations](#)
Michael C. Krygier, Tyler LaBonte, Carianne Martinez, Chance Norris, Krish Sharma, Lincoln N. Collins, Partha P. Mukherjee, and Scott A. Roberts
Nature Communications, 12(1):5414, 2021

WORKSHOP ARTICLES

1. [On the Unreasonable Effectiveness of Last-layer Retraining](#)
John C. Hill, Tyler LaBonte, Xincheng Zhang, and Vidya Muthukumar
ICLR 2025 Workshop on Spurious Correlations and Shortcut Learning
2. [Saving a Split for Last-layer Retraining can Improve Group Robustness without Group Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
ICML 2023 Workshop on Spurious Correlations, Invariance, and Stability
3. [Dropout Disagreement: A Recipe for Group Robustness with Fewer Annotations](#)
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2022 Workshop on Distribution Shifts
4. [Scaling Novel Object Detection with Weakly Supervised Detection Transformers](#)
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
CVPR 2022 Workshop on Transformers in Vision

THESES

1. [Generalization in Dynamic Environments: Foundations and Algorithms](#)
Tyler LaBonte
Doctoral Dissertation, Georgia Institute of Technology, 2026
2. [Finding the Needle in a High-Dimensional Haystack: Oracle Methods for Convex Optimization](#)
Tyler LaBonte
Undergraduate Thesis, University of Southern California, 2021
Winner of the USC Discovery Scholar distinction

Awards

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| 1. ICLR 2026 Top-200 Reviewer (\$675) | 2026 |
| 2. IMS Workshop on Frontiers of Statistical Machine Learning Travel Grant (\$500) | 2025 |
| 3. 2 nd Place Research Talk/Poster Presentation – DoD NDSEG CONFERENCE | 2023 |
| 4. Simons Institute Deep Learning Theory Workshop Travel Grant (\$2,000) | 2022 |
| 5. DoD National Defense Science and Engineering Graduate Fellowship (\$170,000) | 2021 |
| • One of two undergraduates to receive both DoD NDSEG and NSF GRFP in Computer Science | |
| 6. NSF Graduate Research Fellowship (\$138,000—declined) | 2021 |
| 7. USC Discovery Scholar (Research distinction for <100 USC graduates) | 2021 |
| 8. Neo Scholar (Top ~100 CS undergraduates in America) – NEO | 2020 |
| 9. U.S.S. Bowfin Memorial Scholarship (\$5,000) | 2020 |
| 10. 1 st Place Computer Vision Project – TREEHACKS, STANFORD UNIVERSITY | 2019 |

11. 1 st Place Healthcare AI Project – TREEHACKS, STANFORD UNIVERSITY	2019
12. 1 st Place Data Analytics Project – HACKSC, USC	2019
13. Admiral Bernard Clarey Memorial Scholarship (\$7,000)	2018
14. National Top 20 Ethical Hacking Finalist – MAJOR LEAGUE HACKING	2018
15. USC Trustee Scholar (\$250,000)	2017
16. USC Viterbi Fellow (\$24,000)	2017
17. Dolphin Scholarship (\$13,600)	2017
18. Rear Admiral Paul Lacy Memorial Scholarship (\$6,500)	2017
19. National Merit Scholar (\$3,000)	2017

Talks

1. University of Washington School of Computer Science and Engineering – SEATTLE, WA Task Shift: From Classification to Regression in Overparameterized Linear Models	2025
2. Georgia Tech School of Industrial and Systems Engineering – ATLANTA, GA Task Shift: From Classification to Regression via Benign Overfitting	2024
3. Georgia Tech Machine Learning Center – ATLANTA, GA Task Shift: From Classification to Regression via Benign Overfitting	2024
4. Google DeepMind – MOUNTAIN VIEW, CA Towards Last-layer Retraining for Group Robustness with Fewer Annotations	2023
5. Google Cloud Technical Infrastructure – SUNNYVALE, CA Large Language Models for Hardware-Software Code Design	2023
6. DoD NDSEG Conference – SAN ANTONIO, TX Towards Last-layer Retraining for Group Robustness with Fewer Annotations	2023
7. Microsoft Research – REDMOND, WA Weakly Supervised Detection Transformers for Effortless Computer Vision	2021
8. USC Computer Science Theory Group – LOS ANGELES, CA The Distance Oracle for Convex Optimization	2021
9. Google X – MOUNTAIN VIEW, CA Temporal Identity Preservation in Multiple Object Tracking	2020
10. USC Computer Science Theory Group – LOS ANGELES, CA 3D Bayesian CNNs for Credible Geometric Uncertainty	2019
11. USC Center for Artificial Intelligence in Society – LOS ANGELES, CA 3D Bayesian CNNs for Credible Geometric Uncertainty	2019
12. USC Center for Artificial Intelligence in Society – LOS ANGELES, CA Machine Learning Fairness in Word Embeddings	2019

Advising

1. Xincheng Zhang – Georgia Tech MS → Synovus	2024–2025
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| 2. John C. Hill – Georgia Tech BS/MS → Georgia Tech PhD | 2022–2024 |
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Teaching

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| 1. Lecturer/Teaching Assistant (8 lectures) Georgia Institute of Technology
CS 7545: Machine Learning Theory | 2024 |
| 2. Lecturer/Teaching Assistant (12 lectures) Georgia Institute of Technology
CS 7545: Machine Learning Theory | 2023 |
| 3. Undergraduate Teaching Assistant University of Southern California
CSCI 270: Introduction to Algorithms and Theory of Computing | 2021 |
| 4. Instructor USC Center for Artificial Intelligence in Society
Introduction to Machine Learning | 2019 |
| 5. Undergraduate Teaching Assistant University of Southern California
CSCI 170: Discrete Methods in Computer Science | 2018 |

Academic Service

Reviewing

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| 1. NeurIPS | 2023, 2024, 2026 |
| 2. CVPR | 2026 |
| 3. ICLR | 2024, 2026 |
| 4. TMLR | 2025, 2026 |
| 5. CVPR Workshop on Demographic Diversity in Computer Vision | 2025 |
| 6. ICLR Workshop on Spurious Correlations and Shortcut Learning | 2025 |
| 7. ICML | 2025 |

Organizing

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| 1. Organizer, Georgia Tech ML Theory Reading Group | 2021–2023, 2025 |
| 2. System Administrator, Georgia Tech ML Theory GPU Cluster | 2022–2025 |
| 3. Student Organizer, Learning Theory Alliance Mentorship Workshop | 2023 |

Open Source Software

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| 1. Milkshake: Quick and extendable experimentation with classification models
https://github.com/tmlabonte/milkshake | 2023
★ 5 ♪ 3 |
| 2. WS-DETR: Weakly supervised Transformers for scaling novel object detection
https://github.com/tmlabonte/weakly-supervised-detr | 2021–2022
★ 10 ♪ 6 |
| 3. BCNN: 3D Bayesian CNNs for credible geometric uncertainty
https://github.com/sandialabs/bcnn
Transitioned to a production environment by Sandia National Laboratories
19 th most starred Sandia repository out of 693 (March 2025) | 2019–2020
★ 62 ♪ 19 |

4. Tendies: Decoupling deep learning development and deployment 2018
<https://github.com/tmlabonte/tendies> ★ 37 📄 11
Transitioned to a production environment by the Air Force Research Laboratory

Other Activities

1. Fleet Captain, [Georgia Tech Sailing Club](#) 2023–2025
2. House Chair, [USC Hawai'i Club](#) 2020–2021
3. Vice President of Finance, [USC Hawai'i Club](#) 2019–2020